

Decision Theory and Normal-Form Games

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Decision Theory, Briefly

Where We Are

Last Thursday was a write-off, though we had fun talking about some puzzles, so we are folding the essentials of that class into this one.

The compressed version is this section. Everything else from that class is in Chapter 2 of the notes, and I will come back to the parts we need when we need them.

Basic Decision Theory

The basic structure of a decision problem has three parts:

1. Acts
2. States
3. Payoffs

The aim is to identify which acts are worth choosing.

A **game** is going to be a decision problem where the states are the choices of other rational agents.

Payoffs

We assume that the chooser has a **utility function**, usually written u .

This is a function from **outcomes**, i.e., act-state pairs, to **real numbers**.

Some people take these to measure **how good** the outcome is for the chooser; others take them to measure how much the chooser **desires** the outcome.

The simplest approach is to presuppose the view on which these are the same thing, the so-called **preference-satisfaction theory** of welfare. This theory is controversial, to put it mildly, but we'll simply assume it for now. (Later in the course, we'll work through Sen's criticisms of it.)

Utility

The only thing we use utilities for is asking which act has the highest expected utility. Absolute utilities do not do any work, only comparisons between utilities.

If you changed the utility of all outcomes by a constant, *nothing would change*. Utility level **zero** is not of any particular significance.

By default we'll set 0 as the utility of the status quo, but that's a convention with no content.

Every number you see in a payoff matrix is a utility, not a dollar.

Probability

When you set up a decision problem, you have to say what the probabilities are of every state.

I imagine everyone here knows roughly what probabilities are, but to have it on the record, a probability function is a function Pr from propositions to real numbers in $[0, 1]$ satisfying:

$$\text{Pr}(\top) = 1$$

$$\text{Pr}(\perp) = 0$$

$$\text{If } \neg(A \wedge B) \text{ is a theorem then } \text{Pr}(A \vee B) = \text{Pr}(A) + \text{Pr}(B)$$

Conditional Probability

I originally included at this point a long discussion of conditional probability and updating. We will pick up that in a few weeks, when we need them.

Expected Value

Assuming that Pr measures the probability of the states, the **expected value** of each action is given by:

$$E(A) = \sum_{S \in \mathcal{S}} \text{Pr}(S) u(A \wedge S)$$

That is, we sum across the states, in each case multiplying the probability of the state by the utility of performing that action in that state.

Then the one and only rule is **Maximise expected utility!**

Example

Table 1: The umbrella problem again

	Rain (Pr = 0.6)	Dry (Pr = 0.4)
Take Umbrella	6	8
Leave Umbrella	0	10

The expected utility of taking the umbrella is $6 \times 0.6 + 8 \times 0.4 = 6.8$.

The expected utility of leaving the umbrella is $0 \times 0.6 + 10 \times 0.4 = 4$.

Take the umbrella!

What We're Skipping

Everything above is the textbook picture, and it comes with assumptions that are not obviously true.

Two famous arguments say the picture is wrong: the **Allais paradox**, which suggests that not all attitudes to risk are captured by declining marginal utility, and the **Ellsberg paradox**, which suggests that people do not always have a probability for every possibility in their head.

We are not doing these in class this year, and they are not on the quiz. They are in the notes, at 2.7 to 2.9, and they are worth your time.

The point is not that expected utility is wrong; it's that there are interesting arguments that it is wrong, and in a longer course we'd address them.

Normal-Form Games

Simultaneous Move

We're going to start with games that have a distinctive structure.

- ▶ Each player only moves once.
- ▶ These moves are made **simultaneously**.

Examples of this kind of game:

- ▶ Rock, Paper, Scissors
- ▶ Chess between computer programs

Simplifying Assumptions

Those two examples have three things in common that we'll want to eventually drop.

- ▶ They are **two-player games**.
- ▶ They are **zero-sum**.
- ▶ Each player has (more or less) **the same options**

The games we're looking at need not have any of these features.

That said, we will spend a fair bit of time with symmetric two player games.

Payoff Matrix

The simplest way to represent a game is with a **payoff matrix**.

This is sometimes called the **normal form** of the game, and sometimes called the **strategic form**.

Table 2: The Prisoner's Dilemma

	C	D
C	3, 3	0, 5
D	5, 0	1, 1

- ▶ The rows tell us which options player 1 has.
- ▶ The columns tell us which options player 2 has.
- ▶ We'll often call these players **Row** (or just **R**) and **Column** (or just **C**).
- ▶ Each cell has two numbers: First **Player 1's payoff**, then **Player 2's payoff**

Reading the Matrix

	C	D
C	3, 3	0, 5
D	5, 0	1, 1

If Player 1 picks C and Player 2 picks D:

- ▶ Player 1 gets **0**
- ▶ Player 2 gets **5**

Higher numbers are better, by the lights of that player.

These could be something material (e.g., money, years of freedom), or anything the player values (loyalty, equality, etc.)

Payoff Matrix

If we have three players, the usual approach is to have a set of tables, one for each move player 3 could have.

Table 3: P3 does C

	C	D
C	6, 6, 6	3, 7, 3
D	7, 3, 3	4, 4, 0

Table 4: P3 does D

	C	D
C	3, 3, 7	0, 4, 4
D	4, 0, 4	1, 1, 1

We can't do three dimensions on slides, so we have one table for if P3 plays **C**, and another table for if they play **D**.

Formal Elements of a Game

Formally, we define a game by setting three parameters.

1. The number of **Players**.
2. Which **Actions** are available for each player, remembering that this may be different for different players.
3. The **Payoffs** to each player, for each possible combination of actions.

The payoff matrix captures all three at once.

Classic Games

The Prisoner's Dilemma

Two suspects are arrested and interrogated separately.

- ▶ If both **cooperate** (stay silent): light sentence for each.
- ▶ If both **defect** (confess): moderate sentence for each.
- ▶ If one defects and the other cooperates: the defector goes free, the cooperator gets a harsh sentence.

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

What Should You Do?

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

Consider the game from Player 1's perspective.

- ▶ If Player 2 cooperates: I get 3 by cooperating, **5 by defecting**.
- ▶ If Player 2 defects: I get 0 by cooperating, **1 by defecting**.

No matter what Player 2 does, defecting is better for me. The same logic applies to Player 2. So both defect, even though both cooperating would be better for everyone.

The Dilemma

This is why it's called a *dilemma*:

- ▶ Individual rationality leads to (Defect, Defect) with payoffs (1, 1).
- ▶ But (Cooperate, Cooperate) with payoffs (3, 3) is better for *both* players.

This tension between individual and collective rationality is one of the deepest problems in social philosophy.

Coordination Games

Not all games involve conflict. Sometimes the challenge is just to **coordinate**.

Table 5: Which side of the road?

	Left	Right
Left	1, 1	0, 0
Right	0, 0	1, 1

Neither player wants to mismatch. They just need to pick the same side. This is a **pure coordination game**: both players always get the same payoff.

Bach-Stravinsky

What if you want to coordinate, but you *disagree about where*?

Table 6: Bach-Stravinsky

	Bach	Stravinsky
Bach	2, 1	0, 0
Stravinsky	0, 0	1, 2

- ▶ Both prefer going to the same concert to going to different concerts.
- ▶ But Player 1 prefers Bach, and Player 2 prefers Stravinsky.
- ▶ This has elements of both **coordination** and **competition**

Bach-Stravinsky

The older versions of this game have the players be a heterosexual couple with cliched interests.

- ▶ Option **B** is the stereotypical male option, pro wrestling or something.
- ▶ Option **S** is the stereotypical female option; I'll let you guess one.
- ▶ Under this description it was called **Battle of the Sexes**.
- ▶ This more modern name keeps the initials, and the payoff table, but changes the backstory.

The Stag Hunt

There's a game that *looks* like the Prisoner's Dilemma but works very differently.

Table 7: Stag Hunt

	Cooperate	Defect
Cooperate	5, 5	0, 3
Defect	3, 0	3, 3

Two hunters can hunt a stag together (big payoff) or each hunt a hare alone (small but guaranteed payoff). Hunting stag only works if *both* commit to it.

What's Different?

In **Prisoner's Dilemma**, defecting is always better for you, regardless of what the other player does.

In **Stag Hunt**, we have ultimately a coordination game; it's best to do what the other person does.

So why is it hard? Don't we just coordinate on the higher payoff option?

What makes Stag Hunt interesting is that doing the pro-social act is **risky**; if it's 50/50 what the other person will do, it's best to take the individualist option.

Dominant Strategies

What's a Dominated Strategy?

A strategy is **strictly dominated** if there's another strategy that is *always* better than it.

A strategy is **weakly dominated** if there's another strategy that is *always* at least as good as it, and *sometimes* better.

Strategy Rules

Here are two plausible rules (the first, as we'll see, is arguably more plausible):

1. Don't play strictly dominated strategies!
2. Don't play weakly dominated strategies!

We'll focus for now on strict dominance, and come back to weak dominance in a bit, because it's confusing in various ways.

Iterated Elimination of Strictly Dominated Strategies

Consider this game:

Table 8: Iterated dominance example

	L	R
U	1, 1	0, 0
D	0, 3	1, 2

- ▶ No option is dominated for Row.
- ▶ But for Column, they get more playing **L** than **R** no matter what; **L** strictly dominates **R**.
- ▶ And if we know that Column will not play **R**, then on all *remaining* strategies for Column, it's best for Row to choose **U**.

Iterated Elimination

The thought here is that we can sometimes get to a solution of a game by going through and eliminating dominated strategies.

- ▶ Because dominance is defined in terms of quantification over remaining strategies, this elimination can make more strategies dominated.
- ▶ Repeating (or *iterating*) this process sometimes leads to a unique solution.

Iterated Elimination Example: Ice Cream Trucks

Two trucks each choose a spot (1–5) on a beach with 10 customers at each spot.

- ▶ Customers go to the nearest truck; ties split evenly

	1	2	3	4	5
1	25, 25	10, 40	15, 35	20, 30	25, 25
2	40, 10	25, 25	20, 30	25, 25	30, 20
3	35, 15	30, 20	25, 25	30, 20	35, 15
4	30, 20	25, 25	20, 30	25, 25	40, 10
5	25, 25	20, 30	15, 35	10, 40	25, 25

Iterated Elimination Example: Ice Cream Trucks

Step 1: Is spot 1 ever a good choice?

Compare spot 1 vs. spot 2 for Truck 1: no matter where the other truck parks, spot 2 always gets more customers.

Spot 1 is **dominated** by spot 2. By symmetry, spot 5 is dominated by spot 4. Eliminate both spots, for both players.

Iterated Elimination Example: Ice Cream Trucks

After removing spots 1 and 5:

	2	3	4
2	25, 25	20, 30	25, 25
3	30, 20	25, 25	30, 20
4	25, 25	20, 30	25, 25

Step 2: Now spot 2 is dominated by spot 3 (and spot 4 by spot 3). Again, eliminate both spots for both players.

Iterated Elimination Example: Ice Cream Trucks

This game has a unique solution. Both trucks park at **spot 3**.

	3
3	25, 25

A Three Player Version

Table 9: P3 plays F

	L	R
U	2, 3, 3	3, 2, 2
D	5, 1, 4	0, 0, 1

Table 10: P3 plays B

	L	R
U	1, 0, 1	4, 3, 1
D	4, 0, 2	1, 3, 0

This can be solved by first deleting **B** (for back) for P3, then deleting **R** for P2, then deleting **U** for P1. (It's worth verifying all these steps!)

For Next Time

We'll introduce the main solution concept for games that can't be solved by this iterated deletion: **Nash Equilibrium**.

Then we'll come back to the concept of weak domination, and how it introduces various complications.